

Olympiad Test Paper — Science (Class 7) — Paper 3

Chapter 4: Heat

Paper 3 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	4 — Heat
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Designed to punish recall reflexes: hotter-means-more-heat, scale-division misreads, conductor/insulator role-swaps, breeze-direction reversals, and colour-vs-radiation confusions are all baited. Do not write on this paper — mark answers on your answer sheet.

1. A litre of water at 30 °C is poured into an empty pot, and separately a single tablespoon of water at 90 °C is poured into another. A student says the tablespoon 'has more heat because it is hotter.' Which judgement is soundest?

- (A) The tablespoon must contain more heat, since heat content depends only on how hot something is
- (B) Both contain the same heat because heat and temperature always match
- (C) The litre is hotter overall because a larger amount of anything is always warmer
- (D) The tablespoon is at a higher temperature, but the litre at 30 °C can easily contain more total heat energy because it has far more water

2. Two identical metal blocks, one at 80 °C and one at 20 °C, are clamped face-to-face and left wrapped in insulation. After a long time, what is the most accurate description of the final state?

- (A) The hot block stays at 80 °C and the cold block at 20 °C because metal does not share heat
- (B) The cold block becomes hotter than the hot block, since heat keeps flowing into it

(C) Both settle at a common temperature near 50°C , with heat having flowed from the hot block to the cold block until no temperature difference remained

(D) Both blocks cool down to 20°C because heat always moves toward the colder value

3. A child touches a metal railing and a plastic bench that have both sat in the same shaded room overnight. The railing feels much colder. A friend insists the metal 'lost more heat in the night and so is at a lower temperature.' Why is the friend wrong?

(A) The metal really is colder because metals always cool to a lower temperature than plastic

(B) Both objects are at the same room temperature; the metal only feels colder because it conducts heat away from the warm hand faster

(C) The plastic is warmer because plastic generates a little heat of its own

(D) The hand cannot judge plastic, only metal, so the comparison is invalid

4. Equal masses of water at 60°C and cooking oil at 60°C are each left to cool in identical cups in the same room. Which statement about the moment they are first set down is correct?

(A) Heat flows from the oil into the water because oil holds more heat

(B) The water is hotter than the oil because water always carries more heat at a given temperature

(C) They start at the same temperature, so heat will flow from each into the cooler room air

(D) Neither loses heat because both are already at the same temperature as each other

5. A thermometer is dipped into a beaker and the liquid thread stops rising once it reaches a steady mark. What does this steady mark tell us?

(A) The beaker contents have stopped having any temperature

(B) The thermometer's liquid has reached the same temperature as the beaker's contents, and that shared value is the reading

(C) The thermometer has run out of heat and can rise no further

(D) The reading shows how much total heat the beaker contains

6. On a winter night a swimming pool of water and a small metal gate are both at 5°C . A student claims the pool 'feels much hotter than the gate to a thermometer.' What is the correct view?

(A) A thermometer would read 5°C for both; the pool merely stores far more total heat because of its huge amount of water

(B) The thermometer reads a higher temperature in the pool because more water means a higher temperature

(C) The gate reads higher because metal concentrates temperature

(D) Neither has a temperature low enough for a thermometer to read

7. A nurse needs to record a feverish patient's temperature to a tenth of a degree and then carry the thermometer to a desk to read it. Which instrument and feature combination is the right choice?

- (A) A laboratory thermometer, because its wide range is needed for a fever
- (B) A clinical thermometer, but the kink must first be removed so the thread can fall back
- (C) A laboratory thermometer, because only it can be read away from the patient
- (D) A clinical thermometer, because its kink holds the thread in place so the reading does not fall while it is carried away

8. A teacher wants to plot how a beaker of water's temperature changes second by second as it is heated from 10 °C to boiling. Why is a clinical thermometer unsuitable, and what is wrong with the choice?

- (A) It works fine, since a clinical thermometer reads any temperature in Celsius
- (B) It is unsuitable only because it is too short to reach the bottom of the beaker
- (C) Its range stops near 42 °C and its kink prevents the thread from falling, so it cannot track a temperature that rises well above 42 °C or moves up and down freely
- (D) It is unsuitable because its liquid would freeze in hot water

9. A student forgets to shake a clinical thermometer down before placing it under a healthy friend's tongue. The thread had been left holding a previous patient's reading of 39.0 °C. What will the new reading most likely show, and why?

- (A) About 39.0 °C, because the kink keeps the thread from falling, so a healthy 37 °C body cannot pull it back below the old value
- (B) About 37.0 °C, because the thermometer always resets itself when used again
- (C) About 0 °C, because the thread drops to the bottom when not held
- (D) A reading higher than 39.0 °C, because every use adds to the previous reading

10. A laboratory thermometer marked from -10 °C to 110 °C and a clinical thermometer marked 35–42 °C are both available. A cook must check that warm milk for a baby is exactly 38 °C, just barely warm. Which is better and why?

- (A) Only the laboratory thermometer can show 38 °C, since the clinical scale ends at 35
- (B) Only the clinical thermometer can show 38 °C, since 38 is above the laboratory thermometer's range
- (C) Neither can show 38 °C because 38 lies outside both scales
- (D) Either can read 38 °C, but the clinical one is finer for the narrow body-warm range; if continuous monitoring while heating is needed, the laboratory thermometer's free-moving thread suits better

11. Why does a clinical thermometer have a much narrower temperature range than a laboratory thermometer, even though both measure in Celsius?

- (A) Clinical liquids cannot expand much, so the scale must be short
- (B) A narrow range makes the thermometer cheaper, which is the only reason

- (C) Laboratory thermometers measure in different units, forcing a wider scale
(D) Human body temperature only varies over a small band, so a narrow 35–42 °C range can be spread out for fine, accurate fever readings

12. A laboratory thermometer's thread is sitting steadily in a beaker of warm water. If you tip the thermometer almost horizontal to read it more comfortably, what happens to the reading compared with a clinical thermometer in the same situation?

- (A) Both readings stay locked, because all thermometers trap their threads
(B) The laboratory thermometer's reading can change as the thread moves freely, while a clinical thermometer's kink would hold its reading fixed
(C) Both readings change freely, because no thermometer holds a reading
(D) Only the clinical thermometer's reading changes, because it has no kink

13. On a laboratory thermometer the numbered marks are 5 degrees apart and there are 5 equal small divisions between consecutive numbers. The thread rests on the 3rd small division above the 45 mark. What is the temperature?

- (A) 45.3 °C (B) 53 °C
(C) 60 °C (D) 48 °C

14. A clinical thermometer reads 35 to 42, with each one-degree gap split into 5 equal small divisions. The thread stops on the 3rd small division above the 38 mark. What is the reading?

- (A) 38.3 °C (B) 41 °C
(C) 38.06 °C (D) 38.6 °C

15. A laboratory thermometer is marked from –10 °C upward, numbered every 10 degrees with 10 equal divisions between numbers. The thread sits 4 divisions above the –10 mark. What temperature does it show?

- (A) –6 °C (B) –14 °C
(C) –4 °C (D) 4 °C

16. Two laboratory thermometers, A and B, measure the same warm water. A is numbered every 10 degrees with 10 divisions between numbers; B is numbered every 10 degrees with only 5 divisions between numbers. Which can be read more precisely, and by how much per division?

- (A) Thermometer A, because its smallest division is 1 °C versus 2 °C for B
(B) Thermometer B, because fewer divisions mean each one covers a smaller temperature
(C) Both are equally precise, since they cover the same range
(D) Thermometer A, because its smallest division is 0.1 °C versus 1 °C for B

17. A laboratory thermometer reading falls from the 5th division above the 20 mark to the 2nd division above the 20 mark, with each division worth 1 °C. By how much did the temperature drop?

(A) 7 °C

(B) 2 °C

(C) 3 °C

(D) 5 °C

18. A student reads a clinical thermometer where 37 and 38 are one degree apart with 10 equal divisions between them, and writes '37 and 5 divisions = 37.5 °C.' A classmate reading the same thread (5 divisions above 37) writes '37.05 °C.' Who is right and why?

(A) The first student: each division is 0.1 °C, so 5 divisions above 37 is 37.5 °C

(B) The classmate, because 5 small divisions always means hundredths

(C) Both are right, since 37.5 and 37.05 are the same value

(D) Neither, because 5 divisions above 37 is 42 °C

19. A long iron rod has its far end in a flame. Tiny wax pellets are stuck at equal spacing along the rod. As heat travels along the rod, the pellets fall off one after another starting from the flame end. The clearest conclusion is that:

(A) Heat is conducted through the iron from the hot end toward the cold end, melting the wax in sequence as it arrives

(B) The wax pellets each make their own heat once warmed

(C) Hot iron rises and physically carries the heat to the next pellet

(D) The flame's radiation jumps directly to each pellet through the air, ignoring the rod

20. Identical wax-tipped rods of copper, glass, and wood are heated equally at one end. The wax tip melts first on one rod, much later on another, and barely on the third. Which rod loses its wax first, and what does the order reveal?

(A) Wood first, because lighter materials conduct heat fastest

(B) Copper first; the order reveals copper conducts heat fastest, then glass, then wood as the poorest conductor

(C) Glass first, because glass is the best conductor of the three

(D) All three lose the wax at the same instant, since the heat source is identical

21. A metal teaspoon and a plastic stirrer of the same size are both left standing in a mug of hot cocoa. After two minutes you grip the top of each. Which feels hotter at the top, and why?

(A) The plastic stirrer's top, because plastic carries heat upward better

(B) Both tops feel equally hot, because they sat in the same cocoa

(C) The metal spoon's top, because metal conducts the cocoa's heat up to the handle far faster than plastic does

(D) Neither top warms, because heat cannot travel up through a standing object

22. Why does conduction happen readily in solids but very poorly in gases like air?

(A) Gases cannot hold any heat at all, so conduction is impossible

(B) In a solid the particles are packed closely and can pass on vibrations easily, whereas in a gas the particles are far apart and rarely collide, so heat passes very slowly

- (C) Solids are colder than gases, so they conduct better
- (D) Gas particles move too slowly to carry any heat

23. A cook keeps a metal spoon resting in a pot of simmering curry and the handle becomes too hot to grip, while a wooden ladle left the same way stays cool to hold. The best single explanation is:

- (A) Metal conducts heat from the curry along its length to the handle, while wood is a poor conductor and blocks that flow
- (B) The wooden ladle reflects the curry's heat back, keeping its handle cool
- (C) The metal spoon generates extra heat by sitting in the curry
- (D) The curry only heats objects that float, so the wood escapes heating

24. Two rods of the same metal but different thickness are heated equally at one end; the thicker rod warms its far end more slowly per unit cross-section is observed, but a simpler classroom takeaway about conduction is that:

- (A) The thinner rod conducts by radiation while the thicker conducts by conduction
- (B) Both rods conduct heat by the same particle-to-particle process because they are the same material, even if their shapes differ
- (C) Different thicknesses mean different materials, so they conduct by different methods
- (D) Only the thicker rod conducts; the thinner one cannot conduct at all

25. A pan of water is heated only at one small spot on its base. Coloured dye dropped at that spot rises there, spreads across the surface, and sinks down the cooler sides. This pattern is best explained by:

- (A) Conduction carrying the dye straight up through still water
- (B) Convection: heated water expands, becomes lighter and rises, while cooler heavier water sinks to replace it, forming a circulating current
- (C) Radiation pushing the dye in a loop around the pan
- (D) The dye dissolving and moving on its own without any water motion

26. In a room with a heater near the floor, smoke from an incense stick near the heater rises, drifts across the ceiling, and descends along a cool far wall. The smoke is tracing:

- (A) A conduction path, since air conducts heat across the room
- (B) A convection current, because warm air rises and cooler air sinks, circulating around the room
- (C) A radiation beam from the heater to the wall
- (D) Random motion unrelated to the heater's warmth

27. A stove stands on the floor of a tall room. After a while, a thermometer held near the ceiling reads several degrees higher than one held near the floor. Why does the warmth gather near the ceiling?

- (A) Conduction carries the stove's heat straight up through the still air to the ceiling
- (B) Radiation from the stove travels only upward and strikes the ceiling
- (C) The ceiling is naturally warmer than the floor in any room, heated or not
- (D) Air warmed by the stove expands, becomes lighter and rises to collect near the ceiling, while cooler denser air sinks to the floor — a convection current

28. A student argues that convection can carry heat through a solid iron block just as it does through water. Why is this wrong?

- (A) Convection works in iron, but only when the iron is painted black
- (B) Convection works in any material as long as it is hot enough
- (C) Convection needs a fluid whose heated parts can physically move and circulate; the particles in a solid block are fixed in place and cannot flow
- (D) Solids cannot be heated at all, so no transfer can occur in iron

29. Hot water at the top of a tall jar and cold water at the bottom, gently arranged so they do not mix, will set up almost no convection current. Why?

- (A) The warm, lighter water is already on top and the cold, denser water below, so there is no tendency for the layers to rise or sink and circulate
- (B) Because hot water is heavier and wants to stay at the bottom
- (C) Because convection needs the cold water to be on top to start
- (D) Because water can never undergo convection in any arrangement

30. Smoke released over a gently warmed patch of ground on a still day climbs almost straight up rather than spreading sideways. The upward climb is mainly due to:

- (A) Convection: air over the warm ground heats, expands, becomes lighter and rises, carrying the smoke up
- (B) Conduction pushing the smoke upward through the air
- (C) Radiation from the ground lifting the smoke particles
- (D) The smoke being naturally lighter than all air regardless of temperature

31. Sitting beside a campfire on a windless night, you feel warmth on the side of your face that points at the fire, while the air directly above the flames streams upward away from you. What carries the warmth to your face?

- (A) Convection, because the hot air must be blowing onto your face
- (B) Conduction, because the warm air touches your face and passes heat along
- (C) The fire pushing the cold night air aside so warmth can reach you
- (D) Radiation, which travels in straight lines from the fire directly to your skin without needing the air to move toward you

32. An astronaut in the vacuum of space still feels the Sun's warmth on a spacesuit. Which mode of heat transfer reaches across that emptiness, and why can the others not?

- (A) Convection, because hot gases flow through space from the Sun
- (B) Radiation, because it needs no material medium, while conduction and convection both require matter that is absent in a vacuum
- (C) Conduction, because space conducts the Sun's heat directly
- (D) All three reach equally, since space is no different from air

33. Two black metal plates of equal size sit in sunlight: plate X at 300 °C (just been heated) and plate Y at 30 °C. Comparing the heat each radiates away every second:

- (A) Plate Y radiates more, because cooler objects release heat faster
- (B) Plate X radiates much more heat per second, because hotter objects emit more radiation
- (C) Both radiate the same, since they are the same size and colour
- (D) Neither radiates any heat unless placed in shade

34. A thermos flask keeps drinks hot largely because of a near-vacuum gap between its double walls. The vacuum chiefly blocks heat loss by which modes, and why?

- (A) Conduction and convection, because both need matter to carry heat and the vacuum removes the matter between the walls
- (B) Radiation, because radiation cannot cross a vacuum
- (C) Only convection, because conduction still passes freely through a vacuum
- (D) All three, because a vacuum stops every kind of heat transfer completely

35. A dull black surface and a shiny silver surface of equal size are heated to the same temperature and left in a cool dark room. Which loses heat by radiation faster?

- (A) The shiny silver surface, because shiny surfaces radiate heat fastest
- (B) The dull black surface, because dark dull surfaces are better emitters of radiation than shiny ones
- (C) Both lose heat at the same rate, since they are at the same temperature
- (D) Neither radiates in a dark room, because radiation needs light shining on it

36. A student writes three claims: '(1) radiation can travel through a vacuum, (2) conduction can travel through a vacuum, (3) convection can travel through a vacuum.' How many are correct?

- (A) All three
- (B) Only two
- (C) Only one
- (D) None

37. Which single list contains materials that are all good conductors of heat, with no insulator hiding among them?

- (A) Copper, wood, aluminium
- (B) Iron, plastic, silver
- (C) Aluminium, air, copper
- (D) Copper, aluminium, iron

38. A cooking pot has a thick metal base and a fixed plastic handle. Explain the two material choices together: the base must _____ and the handle must _____.

- (A) Resist conducting heat so the food stays cold; conduct heat well so the handle stays warm
- (B) Reflect heat away from the food; absorb heat into the cook's hand
- (C) Store heat without passing it on; release heat into the hand
- (D) Conduct heat well so the food heats quickly; resist conducting heat so the cook's hand is not burned

39. A camper has only an aluminium sheet and a woollen blanket to wrap a hot kettle and keep it warm longer. Which should be wrapped directly around the kettle and why?

- (A) The aluminium sheet, because metal is a poor conductor that traps the heat in
- (B) The aluminium sheet, because metal blocks heat from escaping by sealing it
- (C) The woollen blanket, because wool traps air and is a poor conductor, slowing the kettle's heat loss
- (D) Either works equally, since both feel warm to the touch

40. Why does a foam cup keep a hot drink warmer than a thin metal cup of the same shape?

- (A) Foam is full of trapped air pockets and is a poor conductor, so it slows the drink's heat from escaping, while metal conducts heat away quickly
- (B) Foam produces heat to replace what the drink loses
- (C) Metal is a better insulator than foam, so the metal cup should keep it warmer
- (D) Foam reflects the drink's heat back as light

41. A handle is needed for a hot frying pan. Among silver, copper, aluminium, and wood, which is the correct material for the handle and why?

- (A) Silver, because it is the best conductor and so stays coolest
- (B) Copper, because metals make the most comfortable handles
- (C) Wood, because it is a poor conductor and stays cool enough to grip safely
- (D) Aluminium, because light metals do not conduct heat

42. Why is air, despite being all around us, often described as one of the best insulators when it is trapped in place?

- (A) Its widely spaced particles conduct heat very poorly, so trapped air that cannot circulate barely lets heat through
- (B) Air conducts heat better than metal, which makes it a good insulator
- (C) Air destroys heat, so none can pass through it
- (D) Air is an insulator only when it is moving and circulating freely

43. On a sunny summer afternoon, two people wear shirts identical except for colour: one white, one black. Whose shirt absorbs more of the Sun's radiation, and how does that affect comfort?

- (A) The black shirt absorbs more radiation, so that person feels hotter; the white shirt reflects more sunlight and feels cooler
- (B) The white shirt absorbs more radiation, so the white-shirt wearer feels hotter
- (C) Both absorb the same, so colour makes no difference to comfort
- (D) The black shirt reflects more sunlight, so it feels cooler

44. In a cold but sunny winter market, a vendor stands in direct sunlight and feels warmer in a dark coat than they would in a white one. The reason is that the dark coat:

- (A) Reflects more sunlight, which somehow makes it warmer
- (B) Produces heat from the sunlight it receives
- (C) Absorbs more of the Sun's radiation, warming the wearer in the daytime sun
- (D) Conducts the body's own heat outward faster, warming the wearer

45. Two thermometers are placed inside two identical sealed boxes left in bright sun — one box painted dull black, one shiny white. After an hour, which thermometer reads higher and why?

- (A) The one in the white box, because white surfaces absorb the most sunlight
- (B) Both read the same, because the sun shines equally on both boxes
- (C) The one in the black box, but only because black paint creates heat
- (D) The one in the black box, because the dull black surface absorbs more sunlight and heats the air inside more

46. A roof in a hot desert town is to be painted to keep the rooms below as cool as possible. Should it be painted white or black, and why?

- (A) Black, because dark roofs reflect the most sunlight away
- (B) White, because a light surface reflects most of the Sun's radiation, so less heat is absorbed and the rooms stay cooler
- (C) Black, because dark colours keep buildings cool by absorbing heat
- (D) Either, since roof colour does not affect indoor temperature

47. A driver leaves two cars in the same sunny lot all day, one dark blue and one silver-white, windows closed. Returning, which car's interior is hotter and what mainly causes the difference?

- (A) The dark blue car, because its darker body absorbs more of the Sun's radiation than the light one
- (B) The silver-white car, because pale colours absorb more radiation
- (C) Both the same, because both sat in the same sun for the same time
- (D) The dark blue car, but only because dark paint blocks the breeze

48. At a summer cricket match, two spectators sit side by side in the open sun, one in a white cap and one in a black cap of the same cloth and thickness. After an hour, whose scalp is likely warmer and why?

- (A) The white-cap wearer, because pale colours absorb more sunlight
- (B) Both the same, because the caps are the same cloth and thickness
- (C) The black-cap wearer, because the dark cap absorbs more of the Sun's radiation while the white cap reflects more of it away
- (D) The black-cap wearer, because black cloth makes its own heat in sunlight

49. During a hot day at the coast, a kite flier notices the breeze steadily blowing from the water toward the beach. This daytime sea breeze arises because:

- (A) The sea heats up faster than the land, pushing air toward the beach
- (B) The Sun directly drags the air landward across the water
- (C) The cooler land draws warm sea air inward all day
- (D) The land heats up faster than the sea, so air rises over the warm land and cooler air flows in from over the sea

50. Well after sunset on a calm coast, a sailor finds the wind blowing from the shore out toward the open sea. Why does this land breeze develop at night?

- (A) The land stays warmer than the sea at night, pushing air seaward
- (B) The sea cools faster than the land, so air flows out to sea
- (C) The land cools faster than the sea, so air rises over the warmer sea and cooler air flows out from over the land
- (D) The night air simply reverses every breeze automatically

51. A coastal weather log shows the wind blowing inland at 2 p.m. and out to sea at 2 a.m. The single fact that explains both directions is:

- (A) The wind always switches direction every twelve hours on its own
- (B) By day the land is warmer than the sea, and by night the land is cooler than the sea, reversing which way air rises and flows
- (C) The sea is warmer than the land by day and cooler than the land by night
- (D) The Moon reverses the wind direction at night

52. Sea and land breezes are large-scale examples of which mode of heat transfer, and what does the moving fluid consist of?

- (A) Conduction, with the ground passing heat sideways to the air
- (B) Radiation, with heat beaming from sea to land
- (C) Reflection, with sunlight bouncing between land and sea
- (D) Convection, with air as the moving fluid that rises over the warmer surface and flows in from the cooler one

53. A student claims sea and land breezes can occur even on a coast where the land and sea always stay at exactly the same temperature day and night. Why is this impossible?

- (A) Breezes form from the rotation of the Earth alone, so temperature does not matter
- (B) Breezes form because land and sea heat and cool at different rates, creating a temperature difference; with no difference, air would have no reason to rise on one side and flow in from the other
- (C) Breezes would still form because the sea is always saltier than the land
- (D) Breezes need the land and sea to be at identical temperatures to begin

54. Before sunrise, a fisherman raising a sail finds the wind carrying the boat away from shore and out to sea. Which breeze is this, and what is its cause at that hour?

- (A) A sea breeze, because the land is warmer than the sea before sunrise
- (B) A breeze that needs the Sun already high overhead to blow
- (C) A land breeze caused by the sea cooling faster than the land overnight
- (D) A land breeze, because overnight the land has cooled below the sea's temperature, so air rises over the warmer sea and flows out from the land

55. In the hottest part of summer, doctors advise loose, light-coloured cotton clothing rather than tight dark synthetics. Which combination of reasons best supports this advice?

- (A) Dark loose clothes reflect the most sunlight while trapping cooling air, so dark is better
- (B) Tight clothes keep the body coolest by sealing out all heat
- (C) Light colours reflect more sunlight and loose cotton lets air circulate and sweat evaporate, both keeping the body cooler
- (D) Cotton warms the body by trapping a thick layer of air against the skin

56. Why do loose cotton clothes help keep a person cool in summer heat, beyond just their colour?

- (A) Loose cotton traps a thick still-air layer that keeps body heat in
- (B) Loose cotton produces a cooling chemical when it touches skin
- (C) Loose cotton lets air circulate over the skin and absorbs sweat that then evaporates, both carrying heat away from the body
- (D) Loose cotton reflects body heat back inward to cool the skin

57. Why can two thin sweaters worn together feel warmer than a single sweater of the same total thickness?

- (A) Two sweaters weigh more, and heavier clothing is always warmer
- (B) Two sweaters produce more body heat than one
- (C) The two layers trap an extra pocket of still air between them, and trapped air is a poor conductor that slows heat loss
- (D) Two layers conduct the body's heat outward faster, which feels warm

58. On a freezing morning a sparrow puffs up its feathers and looks rounder. How does this fluffing help it stay warm?

- (A) Fluffing makes the feathers conduct the bird's heat away faster, warming it
- (B) Fluffing causes the bird's body to generate extra heat directly
- (C) Fluffing reflects sunlight onto the bird to warm it
- (D) Fluffing traps a thicker layer of still air among the feathers, and trapped air is a poor conductor that slows the bird's heat loss

59. A blanket is sometimes described as 'warm,' yet a blanket left alone on a cold bed is not hot. How does a woollen blanket keep a sleeping person warm?

- (A) It does not make heat; its trapped air is a poor conductor that slows the escape of the person's own body heat
- (B) The wool generates heat of its own once it is laid on the bed
- (C) The wool conducts the room's cold away from the person
- (D) The wool reflects the cold room air back so it cannot reach the body

60. A jacket stuffed with fluffy down keeps a hiker far warmer than a same-thickness sheet of solid rubber, even though both are bendable. The main reason is that the down jacket:

- (A) Is made of a material that produces heat from the body's sweat
- (B) Holds many pockets of trapped still air, which is a much poorer conductor than solid rubber, so it slows heat loss better
- (C) Conducts the body's heat outward faster, which keeps the hiker warm
- (D) Is heavier than rubber, and heavier clothing is always warmer

Solutions — Science (Class 7), Chapter 4: Heat — Paper 3

Paper 3 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	11	D	21	C	31	D	41	C	51	B
2	C	12	B	22	B	32	B	42	A	52	D
3	B	13	D	23	A	33	B	43	A	53	B
4	C	14	D	24	B	34	A	44	C	54	D
5	B	15	A	25	B	35	B	45	D	55	C
6	A	16	A	26	B	36	C	46	B	56	C
7	D	17	C	27	D	37	D	47	A	57	C
8	C	18	A	28	C	38	D	48	C	58	D
9	A	19	A	29	A	39	C	49	D	59	A
10	D	20	B	30	A	40	A	50	C	60	B

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (D) Heat is the total thermal energy; it depends on both temperature and the amount (mass) of substance. A small amount of very hot water can hold less total heat than a large amount of moderately warm water, so the litre at 30 °C can contain more heat than the tablespoon at 90 °C. The claim that heat depends only on temperature ignores the amount of substance. Heat and temperature are not the same quantity, so they need not match. And a larger amount is not automatically hotter — temperature is set by how vigorously the particles move, not by how much there is.

2. (C) Heat flows from the hotter body to the colder one until both reach the same (equilibrium) temperature; with identical blocks that common value lies between, near 50 °C. Metals readily conduct, so the blocks do not stay separate. Heat stops flowing once temperatures are equal — the cold block cannot overshoot past the hot one. And the hot block warms the cold block, so the pair does not simply drop to the cold value of 20 °C.

3. (B) Objects left together in one room reach the same temperature. Metal feels colder because it is a good conductor and pulls heat from the skin quickly, cooling the touch point fast; the sensation is about heat flow, not a lower temperature. The claim that metal always cools lower ignores thermal equilibrium. Plastic does not produce its own heat. And the hand judges both surfaces by the same rule — the difference is conduction speed, not a flaw special to metal.

4. (C) Both liquids read $60\text{ }^{\circ}\text{C}$, so they are at the same temperature, and each is hotter than the room air; heat therefore flows from each liquid into the cooler air. Heat does not flow between the two cups, since they are not in contact and are at the same temperature anyway. Two things at the same temperature are equally hot regardless of how much heat each stores. And being equal in temperature to each other does not stop them losing heat to the colder surroundings.

5. (B) A thermometer reads correctly only after its bulb liquid reaches the same temperature as what it touches; the thread then holds steady, and that mark is the temperature. The contents always have a temperature — a steady reading does not mean the temperature vanished. The thermometer is not 'out of heat'; it simply matched the sample. And a thermometer shows temperature, not the total heat content, which would also depend on the amount of substance.

6. (A) Temperature does not depend on amount, so a thermometer reads $5\text{ }^{\circ}\text{C}$ in both. The pool holds more heat energy only because it contains so much more water — that is a difference in total heat, not in temperature. More water does not raise the temperature reading. Metal does not 'concentrate' temperature. And $5\text{ }^{\circ}\text{C}$ is well within an ordinary laboratory thermometer's range, so both can be read.

7. (D) Clinical thermometers have a kink that traps the thread so the reading stays fixed once the thermometer is removed, letting the nurse read it later at the desk. A laboratory thermometer has no kink, so its thread would drop as it cooled on the way to the desk, and its range is overkill for a fever. The kink is essential, not something to remove. So the laboratory thermometer is the wrong tool for a reading taken away from the body.

8. (C) A clinical thermometer reads only about $35\text{--}42\text{ }^{\circ}\text{C}$ and its kink locks the thread, so it can neither reach boiling temperatures nor follow a freely changing reading — both fatal for this task. Reading in Celsius does not make it usable beyond its range. The problem is the range and kink, not the length. And the liquid certainly would not freeze in hot water; freezing is not the issue.

9. (A) The kink stops the thread from retreating, so it stays at the trapped $39.0\text{ }^{\circ}\text{C}$; a $37\text{ }^{\circ}\text{C}$ body is not warm enough to push it higher, so the reading stays near $39.0\text{ }^{\circ}\text{C}$ — which is exactly why shaking it down first matters. The thermometer does not reset itself; that is the purpose of shaking. The thread does not fall on its own to $0\text{ }^{\circ}\text{C}$, again because of the

kink. And readings do not accumulate from use to use — the thread simply records the highest temperature reached.

10. (D) 38 °C lies inside both scales (35–42 for clinical, –10 to 110 for laboratory), so both can read it; the clinical gives finer divisions near body warmth, while the laboratory's kink-free thread is handy if you watch the milk rise as it heats. The clinical scale begins at 35, so 38 is within it — it is not below the scale. And 38 is well below 110, so it is inside the laboratory range too. Since both cover 38, 'neither can show it' is false.

11. (D) Because a body's temperature only ranges over a few degrees around 37 °C, the clinical scale can devote its whole length to 35–42 °C, giving fine divisions for precise fever measurement. The liquid expands just as readily; the narrow range is a design choice, not a limit of the liquid. Cost is not the governing reason — accuracy over the body range is. Both thermometers use Celsius, so a unit difference does not explain the wider laboratory scale.

12. (B) A laboratory thermometer has no kink, so its thread moves freely and the reading can shift if conditions or position change; a clinical thermometer's kink locks the thread so its reading is held. Not all thermometers trap their threads — only the clinical kink does. So it is false that both lock or both move freely. And the clinical thermometer, with its kink, is the one that holds its reading steady rather than the one that changes, so the claim that only the clinical reading changes reverses the truth.

13. (D) Five small divisions span 5 degrees, so each small division is $5 \div 5 = 1$ °C. Three divisions above 45 means $45 + 3 = 48$ °C. Reading each division as 0.1 °C (giving 45.3) wrongly assumes ten divisions per 5 degrees. Adding the division count to the next number's region (53) double-counts the gap. And 60 ignores the actual marks entirely.

14. (D) Each one-degree gap is split into 5 equal divisions, so each division is $1 \div 5 = 0.2$ °C. Three divisions above 38 means $38 + 3 \times 0.2 = 38.6$ °C. Treating each division as 0.1 °C (giving 38.3) assumes ten divisions per degree, not five. Adding 3 whole degrees to reach 41 misreads small divisions as whole degrees. And 38.06 misplaces the decimal by a factor of ten.

15. (A) Ten divisions per 10 degrees means each division is 1 °C. Four divisions above –10 means moving up (toward warmer) by 4 °C: $-10 + 4 = -6$ °C. Going the wrong way to –14 treats 'above' as colder. Reading –4 forgets that you start at –10, not –8. And 4 °C ignores the –10 starting mark altogether.

16. (A) A has 10 divisions per 10 degrees, so each = 1 °C; B has 5 divisions per 10 degrees, so each = 2 °C. Smaller per-division steps mean finer reading, so A (1 °C) beats B (2 °C). Fewer divisions actually make each division larger, not smaller, so B is coarser. Equal ranges do not guarantee equal precision — division count matters. And A's smallest division is 1 °C, not 0.1 °C, so claiming a 0.1 °C division for A uses a wrong value.

17. (C) Starting reading: $20 + 5 = 25\text{ }^{\circ}\text{C}$; final reading: $20 + 2 = 22\text{ }^{\circ}\text{C}$; the drop is $25 - 22 = 3\text{ }^{\circ}\text{C}$. Adding the division numbers ($5 + 2$) to get 7 is meaningless. Reporting 2 just copies the final division count. And 5 copies the starting division count — neither is the difference.

18. (A) Ten divisions across one degree make each division $0.1\text{ }^{\circ}\text{C}$, so 5 divisions above 37 is $37 + 0.5 = 37.5\text{ }^{\circ}\text{C}$ — the first student is correct. The classmate's 37.05 places the decimal one digit too far, as if each division were $0.01\text{ }^{\circ}\text{C}$. 37.5 and 37.05 are clearly different numbers. And 42 would require the divisions to be whole degrees, which they are not.

19. (A) In a solid metal rod, heat passes from particle to particle by conduction, reaching and melting the nearer wax first and the farther wax later — exactly the observed order. The wax does not generate heat; it merely melts when heat arrives. Solid iron does not flow upward to carry heat — that bodily movement is convection in fluids, not solids. If radiation alone reached each pellet through air, the rod's order and material would not matter, but the sequence along the rod shows conduction through the metal.

20. (B) Copper is a good conductor and carries heat to its tip fastest, melting that wax first; glass is poorer and wood poorest, so they follow in that order. Being light does not make wood a good conductor — it is actually an insulator. Glass conducts far less than copper, so it is not the best. And an identical heat source does not make poor and good conductors deliver heat at the same instant; the material decides the rate.

21. (C) Metal is a good conductor, so heat from the cocoa travels quickly up the metal spoon to its top, while plastic is an insulator and stays cool at the top — so the metal top feels hotter. Plastic conducts worse, not better, than metal. Sitting in the same cocoa does not equalise the handle temperatures, since the materials conduct at different rates. And heat does travel up a solid by conduction, so 'neither warms' is wrong.

22. (B) Conduction relies on particles passing energy to neighbours; solids have tightly packed particles that hand energy along quickly, while a gas's widely spaced particles seldom touch, making conduction feeble. Gases can hold and carry heat (chiefly by convection), so it is false that they hold none. Solids are not inherently colder than gases — temperature is unrelated to the packing. And gas particles actually move fast, but their large spacing, not slowness, makes conduction poor.

23. (A) Metal is a good conductor, carrying heat from the hot curry up to the handle; wood is an insulator, so little heat reaches its handle and it stays cool. Wood does not work by reflecting heat back into the curry — it simply conducts poorly. The metal spoon does not create heat; it only transmits the curry's heat. And whether an object floats has nothing to do with conduction along its handle.

24. (B) Conduction is the passing of heat from particle to particle, and since both rods are the same metal, both transfer heat by this same process regardless of thickness. Thickness

does not switch the mechanism to radiation. Different thickness does not mean different material — they are stated to be the same metal. And a thin rod still conducts; being thin does not stop conduction.

25. (B) When water is heated it expands, grows less dense, and rises; cooler denser water sinks to take its place, and this circulation (convection) carries the dye in the looping path described. Conduction does not move the water bodily, so it cannot trace a loop. Radiation does not drive fluid loops within the pan. And the dye is a passive marker — it moves because the water moves, not by itself.

26. (B) The heater warms nearby air, which expands, rises, travels along the ceiling, cools at the far wall and sinks — a convection loop, which the smoke makes visible. Air conducts heat very poorly, so the path is not conduction. Radiation travels in straight lines and does not loop around a room. And the smoke's organised circuit is driven by the heater's warmth, so it is not random.

27. (D) Air warmed by the stove expands, becomes less dense and rises to collect near the ceiling, while cooler denser air sinks to the floor — a convection current that makes the ceiling thermometer read higher. Conduction through still air is far too slow to carry the stove's heat straight up. Radiation spreads out in all straight-line directions, not only upward, so it would not preferentially pool warmth at the ceiling. And a ceiling is not 'naturally' warmer than a floor in an unheated room — the gradient here is created by the stove's convection.

28. (C) Convection requires a fluid — a liquid or gas — whose warmed portions can rise and circulate; in a solid the particles are locked in fixed positions and cannot flow, so heat moves through iron by conduction instead. Paint colour affects radiation, not whether convection can occur in a solid. Heat alone cannot make a rigid solid's particles circulate. And solids certainly can be heated — they just transfer heat by conduction, not convection.

29. (A) Convection is driven by warm fluid rising and cold sinking; here the warm water is already on top and the cold below, the stable arrangement, so there is nothing to drive circulation. Hot water is lighter, not heavier, so it does not sit at the bottom. Convection starts when heating happens at the bottom (cold on top, warm below), not the reverse. And water clearly can convect — just not in this already-settled arrangement.

30. (A) The warm ground heats the air just above it; that air expands, becomes less dense and rises as a convection current, lifting the smoke. Conduction does not bodily push air upward. Radiation travels in straight rays and does not lift particles by buoyancy. And the smoke rises here because the surrounding air is heated and rising, not because smoke is inherently lighter than all air.

31. (D) Radiation travels in straight lines and needs no medium, so heat reaches the face that points at the fire even though the hot air rises straight up and not toward you.

Convection here carries hot air upward, away from your face, so it cannot be the cause. Air is a poor conductor and the gap is large, so conduction is negligible. And the fire does not warm you by shoving cold air aside — the direct radiation does the warming.

32. (B) Only radiation can cross a vacuum, since it requires no medium; conduction needs touching particles and convection needs a flowing fluid, neither of which exists in empty space. There is no continuous gas filling space to carry convection from the Sun. There is no matter in a vacuum to conduct through. And space, being empty, is very different from air, so the three modes do not act equally there.

33. (B) All warm objects radiate, and the hotter an object the more heat it emits each second, so plate X at 300 °C radiates far more than plate Y at 30 °C. Cooler objects radiate less, not more, so plate Y does not win. Equal size and colour do not make their emission equal when their temperatures differ so much. And objects radiate whether in sun or shade — shade is not required.

34. (A) A vacuum has no particles, so it cannot support conduction (needs touching particles) or convection (needs a flowing fluid); removing the matter between the walls blocks both. Radiation actually does cross a vacuum, so the gap does not stop it — that is why the inner surfaces are also shiny. Conduction cannot pass through a vacuum, so the 'only convection' claim is wrong. And the vacuum does not stop radiation, so it does not block all three.

35. (B) Dull, dark surfaces are good emitters of radiation, so the black surface radiates heat away faster than the shiny silver one at the same temperature. Shiny surfaces are poor emitters, so silver loses heat more slowly, not faster. Equal temperature does not make their emission equal, because surface type matters. And objects radiate their own heat regardless of any light shining on them, so a dark room does not stop emission.

36. (C) Radiation needs no medium and crosses a vacuum, so claim 1 is correct. Conduction needs touching particles and convection needs a flowing fluid; a vacuum has neither, so claims 2 and 3 are false. That leaves exactly one correct claim. 'All three' wrongly grants conduction and convection a vacuum path. 'Only two' wrongly counts one of conduction or convection as working. 'None' wrongly denies radiation, which truly does cross a vacuum.

37. (D) Copper, aluminium and iron are all metals and good conductors, so this list is all conductors. The list with copper, wood and aluminium hides wood, which is an insulator. The list with iron, plastic and silver hides plastic, an insulator. And the list with aluminium, air and copper hides air, which is one of the best insulators. Only the all-metal list is purely conductors.

38. (D) The base is metal (a good conductor) so it passes the stove's heat into the food quickly, while the handle is plastic (an insulator) so it resists conducting heat and stays

cool enough to hold safely. Reversing the roles — base resisting, handle conducting — would leave food cold and burn the hand, the opposite of the design. The base does not reflect heat away from food; it conducts heat into it. And the design is about conducting versus insulating, not storing heat then releasing it into the hand.

39. (C) Wool traps air and is a poor conductor, so wrapping it around the kettle greatly slows heat from escaping. Aluminium is a good conductor, so it would carry the kettle's heat away to its outer surface quickly — the opposite of what is wanted. Metal does not 'seal in' heat by blocking it; it conducts heat through itself. And the two materials are not equally effective, since wool insulates far better than metal.

40. (A) Foam contains many trapped air pockets and is a poor conductor, so heat escapes slowly; a thin metal cup conducts the drink's heat outward fast, cooling it sooner. Foam does not generate heat. Metal is a good conductor, not a better insulator than foam, so the metal cup actually cools faster. And foam does not work by reflecting heat back as light — it works by being a poor conductor with trapped air.

41. (C) A handle must be a poor conductor so it stays cool; wood is an insulator and is the right choice. Silver is the best conductor of the four and would get hot fastest — a good conductor makes a bad handle. Copper is also a metal and conducts heat well, so it would heat up. And aluminium, though light, is still a metal and conducts heat well, so 'light metals do not conduct' is false.

42. (A) Air's particles are far apart and conduct heat very poorly, so when it is trapped and cannot circulate it lets almost no heat through — making trapped air an excellent insulator. Air conducts far worse than metal, not better. Air does not destroy heat; it simply transmits it slowly. And trapped, still air insulates best — moving air would carry heat away by convection, the opposite of insulating.

43. (A) Dark colours absorb more radiation and light colours reflect more, so the black shirt heats up more and the white shirt stays cooler — which is why white is preferred in summer. White reflects rather than absorbs, so the white wearer does not get hotter. Colour clearly does affect absorption, so the two are not the same. And black absorbs rather than reflects, so it does not feel cooler.

44. (C) Dark colours absorb more of the Sun's radiation, so in direct winter sun a dark coat warms the wearer more than a white one would — useful in cold weather. Reflecting sunlight would keep the wearer cooler, not warmer, so reflection cannot be the reason. The coat does not manufacture heat; it absorbs the Sun's. And conducting body heat outward faster would cool the wearer, the opposite of warming.

45. (D) A dull black surface absorbs more of the Sun's radiation, so the black box warms its inside air more and its thermometer reads higher. White reflects most sunlight, so it absorbs least and stays cooler — it does not absorb the most. Equal sunshine does not give

equal readings, because the surfaces absorb different fractions. And black paint does not 'create' heat; it absorbs the Sun's radiation.

46. (B) A white roof reflects most of the Sun's radiation, absorbing little, so the rooms stay cooler — the right choice in a hot place. Black reflects little and absorbs much, so it does not reflect sunlight away. Absorbing heat warms a building, so a black roof would make rooms hotter, not cooler. And colour does affect indoor temperature through how much radiation is absorbed, so 'either' is wrong.

47. (A) Darker surfaces absorb more solar radiation, so the dark blue car's interior heats up more than the silver-white one's. Pale colours reflect rather than absorb, so the white car absorbs less, not more. Equal sun and time do not equalise them, because the colours absorb different amounts. And the difference is from absorbed radiation, not from paint blocking a breeze.

48. (C) Dark colours absorb more of the Sun's radiation and light colours reflect more, so the black cap heats up more and warms the scalp beneath it, while the white cap stays cooler. Pale colours reflect rather than absorb, so the white cap does not warm more. Same cloth and thickness do not make them equal, because colour decides how much radiation is absorbed. And black cloth does not generate its own heat — it simply absorbs the Sun's radiation.

49. (D) Land heats faster than water in the day, so air over the warm land rises and cooler sea air flows in to replace it — that inflow is the sea breeze. The sea actually heats more slowly than land, so it is not the sea heating fastest. The Sun does not physically drag air sideways. And the land is the warmer surface in daytime, not the cooler one, so it does not 'draw in' as if cool.

50. (C) At night land cools faster than the sea, so the sea stays warmer; air rises over the warm sea and cooler air flows out from the cooled land — that outflow is the land breeze. The land is actually cooler than the sea at night, not warmer. The sea cools more slowly than land, so it is not the sea cooling fastest. And the reversal is caused by the day-night temperature swap between land and sea, not by an automatic rule.

51. (B) Land heats faster by day (so it is warmer, air rises there, breeze blows inland) and cools faster by night (so it is cooler, air rises over the warmer sea, breeze blows seaward) — one day-night temperature swap explains both. There is no automatic twelve-hour switch rule; the temperatures cause it. The sea actually stays cooler by day and warmer by night, the reverse of the claim that the sea is warmer by day and cooler by night. And the Moon does not reverse winds; uneven land-sea heating does.

52. (D) Breezes are air moving because of uneven heating — warm air rises and cooler air flows in to replace it — which is convection on a coastal scale, with air as the moving fluid. Conduction does not move air bodily across the coast. Radiation is straight-line heat from

a source, not the circulation of air. And reflection of sunlight is not a mode of heat transfer that creates winds.

53. (B) The driving cause of these breezes is the temperature difference from land and sea heating and cooling at different rates; with no difference, no side is warmer to make air rise, so no breeze forms. Earth's rotation does not by itself create a local sea breeze. Saltiness does not drive these breezes — the temperature difference does. And identical temperatures would remove the difference that breezes need, so they cannot be a requirement for the breeze.

54. (D) Before sunrise the land has cooled below the sea overnight, so the sea is warmer; air rises over the warm sea and cool air flows out from the land toward the sea — a land breeze. The land is not warmer than the sea before sunrise, so it is not a sea breeze. The breeze does not require the Sun overhead; it works at night from stored daytime warmth differences. And it is the land, not the sea, that cools faster overnight, so blaming the sea's faster cooling names the wrong surface.

55. (C) Two effects help: light colours reflect sunlight so less heat is absorbed, and loose cotton lets air flow and sweat evaporate, carrying heat away — together keeping you cool. Dark clothes absorb, not reflect, sunlight, so dark is worse in summer. Tight clothes restrict airflow and sweat evaporation, so they do not keep you coolest. And in summer we want to lose heat; trapping a warm air layer (what wool does in winter) is the opposite of the goal.

56. (C) Loose cotton allows air to move over the skin and soaks up sweat which then evaporates; both processes carry heat away and cool the body. Trapping a still-air layer would keep heat in (good for winter wool, bad for summer cooling), so that is the wrong mechanism. Cotton does not make a cooling chemical. And reflecting body heat 'back inward' would warm the skin, not cool it.

57. (C) Wearing two layers traps an extra cushion of still air between them, and trapped air is a poor conductor, so it slows heat escaping and feels warmer than one layer of the same thickness. Weight does not determine warmth — trapped air does. Clothes do not generate body heat. And conducting heat outward faster would cool you, not warm you, so that reverses the effect.

58. (D) Puffed feathers trap more still air, and trapped air is a poor conductor, so the bird loses heat more slowly and stays warm. Conducting heat away faster would cool the bird, not warm it. Fluffing rearranges feathers; it does not make the body produce extra heat. And the warmth comes from trapped insulating air, not from reflecting sunlight (which would not work at all on a cloudy freezing morning).

59. (A) Wool traps air, a poor conductor, so it slows the loss of the body's own heat — the person stays warm because their heat is kept in, not because the blanket is a heat source.

Wool does not generate heat, which is why it is not hot on its own. 'Conducting cold away' is not how warmth works — there is no flow of cold, only heat moving from warm to cool. And the blanket works by slowing heat loss, not by reflecting cold air.

60. (B) Down holds many pockets of still trapped air, and trapped air conducts heat very poorly, so a down jacket insulates better than solid rubber of the same thickness. The jacket does not produce heat from sweat. Conducting body heat outward faster would cool the hiker, the opposite of staying warm. And weight does not determine warmth; the trapped air does — down is actually lighter than rubber yet warmer.